

Mounika Ayyapu | Data Analyst

Chicago, IL | +1 (312)-478-3132 | mounikaayyapuu@gmail.com | [linkedin.com/in/mounika-ayyapu](https://www.linkedin.com/in/mounika-ayyapu)

PROFESSIONAL SUMMARY

Data Analyst with over 4 years of experience driving **data-driven decision-making**, business intelligence, and process optimization across healthcare, retail, and financial domains. Proficient in **SQL, Python, Power BI, Tableau, and cloud platforms** (Azure, AWS, Snowflake) to extract, transform, and visualize complex datasets, delivering actionable insights that improve operational efficiency and reduce manual effort. Experienced in designing **interactive dashboards, predictive models, and scalable ETL pipelines**, ensuring high data quality, governance, and compliance. Adept at collaborating with cross-functional teams to translate business requirements into analytics solutions, streamline reporting workflows, and support strategic initiatives. Skilled in **statistical analysis, forecasting, clustering, and trend modeling**, with a proven ability to turn data into measurable business impact.

SKILLS

Data Analysis & Querying: SQL (T-SQL, PostgreSQL, Redshift), Advanced Excel (PivotTables, VLOOKUP, INDEX-MATCH, Macros, Conditional Formatting), Data Wrangling, Exploratory Data Analysis (EDA), Schema Mapping, Data Validation, Anomaly Detection, Aggregation, Segmentation, Performance Benchmarking

Programming & Scripting: Python (pandas, NumPy, matplotlib, seaborn, scikit-learn), PySpark, Shell Scripting, Automation Scripting, Data Cleaning, Jupyter Notebook

Business Intelligence & Visualization: Power BI (DAX, Power Query, Data Modeling, KPI Dashboards), Tableau, Excel Dashboards, Structured Reporting Packages, Interactive Dashboards, Data Storytelling, Visual Analytics, Performance Monitoring

Cloud & Data Platforms: Azure (Data Factory, Synapse, Data Lake), AWS (S3, Redshift, SageMaker), Snowflake, Cloud Data Storage, Cloud Data Integration, Cloud Pipeline Optimization, Enterprise-Level Data Architecture

ETL & Data Orchestration: ETL Pipeline Design, SQL Based Transformations, Apache Airflow, Azure Data Factory, Pipeline Automation, Pipeline Monitoring, Data Quality Validation, Data Integration from Heterogeneous Sources

Statistical & Predictive Analytics: Time Series Forecasting, Regression Analysis, Churn Modeling, A/B Testing, Variance Analysis, Trend Modeling, Predictive Modeling, Classification and Regression Techniques, Fraud Detection, Market Segmentation, Insights Generation

Healthcare & Compliance Analytics: Claims Data Processing, Readmission Analysis, Patient Flow Optimization, KPI Monitoring, HIPAA Compliant Reporting, RBAC-Based Data Governance, Healthcare Data Quality Checks, Auditing, Clinical and Operational Analytics

Collaboration, Tools & Methodologies: Git, GitHub, Jira, Confluence, ServiceNow, Agile/Scrum Methodologies, Stakeholder Communication, Cross-Functional Data Delivery, SOP Documentation, Reporting Standards, SDLC Alignment, Team Collaboration

WORK EXPERIENCE

Data Analyst Intern | UnitedHealth Group | USA **Dec 2024 – May 2025**

- Conducted comprehensive quantitative and qualitative analysis on complex healthcare datasets using **SQL and Excel**, identifying **patterns, correlations, and anomalies** that enabled business teams to make informed, data-driven operational improvements, resulting in a **15% increase in reporting efficiency and faster patient outcome analysis**.
- Planned and implemented interactive dashboards and computerized reporting tools in Power BI and Tableau, allowing leadership and client stakeholders to monitor KPIs, visualize trends in real-time, and make faster, more accurate strategic decisions, which **reduced manual reporting time by 25%**.
- Assisted in innovation and research initiatives across product, process, and technology domains, producing detailed analytical findings, structured documentation, and exploratory models that guided senior analysts in evaluating opportunities, **leading to actionable recommendations adopted in ongoing projects**.
- Executed thorough data audits, validations, and integrity checks across multiple platforms, identifying discrepancies and implementing corrective measures that strengthened governance, refined reporting accuracy, and ensured compliance with regulatory standards, **enhancing data reliability for 100% of monitored datasets**.
- Partnered with mentors, project managers, and cross-functional stakeholders to translate complex business requirements into analytics solutions, gaining hands-on experience in healthcare workflows and contributing to collaborative decision-making, which **accelerated project completion timelines by 10%**.
- Delivered presentations of analytical findings and dashboards to both technical and non-technical audiences, refining communication skills while ensuring critical insights were clearly understood, **enabling leadership to make timely operational decisions**.
- Expanded automation scripts and reporting utilities that streamlined repetitive tasks, reducing manual effort by 20% and freeing analysts to focus on higher-value activities, **improving team productivity and accelerating insight delivery**.

Data Analyst | Tata Consultancy Services | India **Jul 2021 – Jul 2023**

- Collected, cleaned, and validated **large datasets** from multiple structured and unstructured sources using **SQL, Excel, and data catalog tools**, ensuring consistency and reliability for reporting, analytics, and business intelligence initiatives.
- Performed exploratory data analysis, statistical modeling, and predictive analytics with Python and R, discovering actionable trends and patterns that informed strategic decisions, **contributing to a 20% increase in ROI for targeted campaigns**.
- Built, optimized, and sustained dynamic dashboards, automated reports, and visual analytics solutions in Power BI and Tableau, providing executives with interactive tools to track KPIs, **enhancing decision-making speed and operational transparency**.
- Collaborated with business leaders and client stakeholders to define KPIs, structure reporting frameworks, and deliver insights, ensuring analytics outputs aligned with organizational goals, **leading to improved alignment between business and analytics teams**.
- Resolved complex data inconsistencies and workflow challenges through targeted investigation and proactive recommendations, improving efficiency and strengthening data management practices, **resulting in a 25% reduction in reporting issues**.
- Created and deployed scalable ETL pipelines that integrated heterogeneous systems into centralized repositories, improving data availability and supporting predictive modeling, **accelerating data delivery timelines by 15%**.
- Implemented process improvement strategies for reporting, collection, and analysis workflows, reducing redundancy, enhancing governance, and creating reliable mechanisms for ongoing data-driven decision-making, **boosting overall efficiency by 18%**.

Data Analyst | Hexaware Technologies | India **May 2019 – Jun 2021**

- Extracted, transformed, and maintained **large datasets** using advanced **SQL queries**, integrating data from multiple systems to create consistent pipelines and deliver accurate information for enterprise reporting and analytics.
- Performed data cleansing, anomaly detection, and preprocessing routines to ensure datasets were accurate, complete, and compliant with regulatory standards, **improving data trustworthiness for business intelligence initiatives**.
- Applied statistical analysis, clustering methods, and predictive modeling with Python and R, generating actionable insights, optimized business processes, and supported strategic planning, **leading to a 15% improvement in operational efficiency**.
- Developed and deployed interactive dashboards, automated reports, and visual analytics in Tableau and Power BI to monitor performance metrics, detect patterns, and respond proactively, **reducing manual reporting effort by 25%**.
- Partnered with engineering, product, and business teams to translate complex requirements into scalable analytics solutions, ensuring deliverables addressed key objectives, **resulting in measurable business value and refined project outcomes**.
- Authored comprehensive technical documentation, workflow guides, and validation processes, supporting testing, deployment, and adoption, **minimizing operational risks and ensuring smooth project execution**.
- Designed forecasting models and trend analysis frameworks for sales and operational data, enabling accurate demand planning, resource optimization, and cost reduction, **achieving a 10% decrease in operational costs while improving forecast accuracy**.

PROJECTS

Intelligent Document Search and Retrieval Platform

- Built a **robust indexing framework** that processed and organized more than **1,000 unstructured documents**, enabling employees to quickly locate relevant files and reducing time spent on manual data discovery.
- Applied **semantic similarity models and clustering algorithms** to deliver **context-aware recommendations**, helping users navigate related documents efficiently while uncovering deeper connections within large-scale datasets.
- Architected the solution on a **scalable infrastructure**, ensuring that the platform maintained strong performance, minimized latency, and supported enterprise-level workloads without compromising reliability or speed.
- Boosted organizational productivity by **accelerating decision-making** through intelligent, precise, and accessible document retrieval, allowing business teams to make informed choices backed by accurate and contextual insights.
- Introduced a **comprehensive monitoring system** that tracked query accuracy, search response time, and user interaction patterns, providing continuous feedback loops to enhance platform performance and ensure long-term sustainability.

Retail Consumer Insights and Market Segmentation

- Analyzed over **100,000 retail transactions** to identify consumer behavior trends, seasonal purchasing shifts, and top-selling categories, enabling retailers to optimize product placement and refine inventory strategies effectively.
- Utilized **K-Means clustering and unsupervised learning** to segment customers into meaningful groups, which allowed the marketing team to design targeted campaigns that boosted personalization and generated a **25% ROI uplift**.
- Initiated **interactive dashboards** in Power BI and Tableau, providing executives with real-time visibility into KPIs such as churn probability, customer lifetime value, and product-level sales performance across geographies.
- Partnered with **marketing and sales teams** to translate complex analytical results into actionable initiatives, ensuring that data-driven strategies aligned with promotional planning and customer engagement goals.
- Enhanced overall **business decision-making processes** by delivering timely insights, improving campaign accuracy, boosting profitability, and establishing a culture of analytics-driven retail strategy.

EDUCATION

- **Masters in Computer Science** May 2025
Illinois Institute of Technology | Chicago, IL.
- **Bachelors in Information Technology** May 2021
Sree Vidyanikethan Engineering College | AP, India.

CERTIFICATIONS

Data Visualization – Excelerate
Microsoft Certified: Power BI Data Analyst Associate
Joy of computing using python – NPTE